

AI–Based Benchmarking of the Methane Metauniverse (MMU)

A Comparative Audit Using Gemini, Copilot, and Independent Evaluation

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Abstract

We report a rule-bound comparative benchmark of the Methane Metauniverse (MMU), a corpus-based geometric-elastic framework developed under a fixed Canon. Three audit tracks were performed: Google Gemini, Microsoft Copilot, and an independent conservative consolidation. The aim is not advocacy or empirical validation, but a reproducible epistemic assessment: whether an external AI system can reconstruct the MMU from public OSF/ResearchGate documentation, formulate technically meaningful objections, and classify the framework relative to established reference theories under identical scoring criteria. All audits were constrained by controlled source access, explicit traceability requirements, and a fixed five-axis score (Understanding, Consistency, Parsimony, Explanatory Power, Falsifiability). We document differences in auditor access and scoring strictness and provide a conservative consolidated result. The benchmark indicates that MMU is structurally coherent and highly compressed in assumptions, while remaining formally incomplete in a small set of key closures (notably projection normalization, consolidated operator/PDE derivations, and fully specified kill-test protocols). The study is presented as a neutral pre-review filter to reduce informational load prior to expert human peer review.

1 Introduction

This work explores whether a tightly controlled AI–human collaboration can contribute to the construction of a physically coherent and radically non-standard theoretical framework within a limited time span. Over approximately three months, a new geometric-elastic model of fundamental physics, referred to as the Methane Metauniverse (MMU), was developed under continuous human guidance, explicit theoretical constraints, and systematic documentation. The physical foundations of the framework are consolidated in a fixed canonical reference [36].

The objective of this study is not to establish the physical correctness of the MMU framework. Instead, it addresses a methodological question of growing relevance in AI-assisted science: whether AI augmentation can accelerate early-stage theory formation while preserving conceptual discipline, internal consistency, and anchoring to established physical principles [35].

Beyond model construction, the approach explicitly tests whether the resulting framework is sufficiently developed and documented to be examined independently by external researchers. In particular, the study probes whether the MMU corpus, when accessed solely through public documentation, provides enough structural clarity, definitional precision, and contextual separation to allow meaningful reconstruction, critique, and comparative evaluation without reliance on internal development knowledge.

A central aspect of the investigation is therefore whether a model developed in this manner can be taught to previously untrained AI systems using only publicly available documentation, and whether these systems are subsequently able to critically evaluate the model relative to

established physical theories. This includes assessing whether independent AI systems converge on similar interpretations, identify comparable open problems, and remain anchored to well-tested physical facts rather than introducing speculative extensions or category errors [34, 35].

The MMU framework is intentionally broad in scope, addressing multiple physical domains including spacetime structure, quantum phenomena, particle properties, and effective gravitational behavior [1, 9, 14, 20, 21]. As a consequence, its development and evaluation cannot be confined to a single comprehensive paper. Instead, the framework is presented as a modular corpus of interconnected publications, each addressing a specific aspect of the model, while remaining bound to a shared Canon that defines assumptions, non-assumptions, projection principles, and scaling laws [36].

This work evaluates whether such a corpus-based approach, supported by AI augmentation and explicit epistemic controls, can yield a theoretical program that is not only internally coherent but also externally auditable. The emphasis is placed on openness of information, reproducibility, and auditability as necessary conditions for scientific seriousness at this early, speculative stage of development.

2 Relation to Prior AI-Based Audit Study

The present work builds directly on a prior methodological study investigating AI-augmented theory formation in the context of the MMU framework [35]. In that earlier study, independent AI systems were employed as epistemic auditors under a fixed protocol, without targeted training or adaptation. The objective was to assess whether untrained AI systems could reconstruct, critique, and classify a non-standard theoretical framework solely on the basis of public documentation.

In that setting, the auditing AI systems were intentionally not guided toward specific theoretical benchmarks. Instead, they evaluated the MMU framework against general scientific criteria such as internal consistency, falsifiability, and conceptual clarity. The focus was on reconstructability and epistemic discipline rather than on comparative placement within the landscape of established physical theories.

Under these constraints, the MMU framework was classified as a *structurally coherent but formally incomplete speculative physical theory* [35]. The evaluation converged across internal and external AI audits and identified the same primary limitations, most notably the absence of appendix-grade derivations for key operators and spectra, the lack of fully worked numerical predictions with explicit uncertainty propagation, and the incompleteness of experimentally decisive falsification protocols.

The present study extends this approach by shifting the emphasis from reconstruction to comparative stress testing. Here, the MMU framework is examined under conditions that explicitly probe whether its scientific structure is sufficiently developed to allow meaningful comparison with existing theoretical programs. The goal is not to establish superiority or correctness, but to evaluate whether MMU can be positioned coherently relative to established frameworks when assessed under equivalent methodological constraints.

In this context, AI systems are again not trained on the MMU in an adaptive or optimizing sense. Instead, they are exposed to the public MMU corpus under the same fixed standards applied to other theories. This allows the audits to function as stress tests of scientific structure, revealing whether MMU exhibits comparable levels of conceptual organization, explicit assumptions, and vulnerability to falsification.

This comparative perspective transforms the role of AI from that of a reconstruction assistant to that of a neutral structural probe. The analysis therefore complements the earlier audit study by moving from the question of whether MMU can be understood at all to the question of whether it can be taken seriously as a scientific research program alongside existing theories.

3 Methodology and Development Process

The MMU framework was developed iteratively under continuous human supervision and explicit conceptual guidance. Development proceeded in small, controlled steps, with emphasis on internal coherence, dimensional consistency, and a strict separation between assumptions, derived results, and open problems. All physical statements were required to remain consistent with a fixed canonical reference defining the scope and interpretation of the model [36].

A substantial portion of the effort was devoted not to introducing new physical claims, but to constructing a coherent documentation and control architecture. This included the formulation and stabilization of the Canon, the development of a structured corpus of modular publications addressing distinct physical domains [1, 9, 14, 20, 21], and the creation of a retrieval-augmented knowledge base designed to anchor AI reasoning to explicit sources. In parallel, strict instructional constraints were imposed to suppress conceptual drift, silent redefinition of terms, and hallucination effects in AI-assisted analysis [35].

Once a stable reference state was reached, independent AI systems with no prior exposure to the MMU project were engaged. These systems were restricted to publicly available documentation hosted on OSF and ResearchGate and were tasked with reconstructing the framework, identifying unresolved issues, and formulating potentially falsifying criticisms following a predefined audit protocol [34, 35].

The audits were explicitly not designed to verify or validate the physical claims of the MMU. Instead, they function as neutral probes of epistemic maturity, evaluating whether the resulting framework is internally consistent, physically coherent, and scientifically serious in its formulation. Particular attention was paid to whether independent AI systems converge on similar interpretations and critiques when constrained to the same public information.

In this context, auditability, openness of information, and reproducibility were treated as primary quality criteria. The ability of independent AI systems to reconstruct and critically assess the model using only public sources is interpreted as an indicator of documentation quality and conceptual clarity, rather than as evidence of physical correctness or empirical confirmation.

4 Benchmark Protocol

The comparative evaluation of the MMU framework was conducted using a fixed, normative benchmark protocol defined prior to the audits. The protocol was designed to ensure symmetry across all evaluated theories, reproducibility of results, and explicit separation between physical content and sociological factors. A complete specification of the protocol is provided in Appendix A.

4.1 Normative Constraints

All audits were required to respect the following binding rules, applied uniformly to the MMU framework and to all reference theories:

- **Canon First:** Canon Version 2.0 serves as the normative reference defining assumptions, non-assumptions, definitions, projection principles, and scaling laws for the MMU framework [36].
- **Corpus-Assisted Evaluation:** Canon-referenced MMU publications archived on OSF and ResearchGate may be used for explicit derivations, definitions, and numerical results, provided all references are explicitly cited [1, 21, 35].
- **No Sociological Bias:** Acceptance status, popularity, citation counts, or community consensus are explicitly excluded from all scoring criteria.

- **Traceability:** Every evaluative claim must be traceable either to the Canon or to a specific, explicitly cited MMU publication. Unsupported or implicit assumptions are not permitted.

4.2 Evaluation Criteria

Each theory was evaluated using an identical set of five epistemic criteria, applied with the same strictness and interpretation across all models:

1. **Understanding (U):** Clarity of definitions, scope, and explicit assumptions.
2. **Internal Consistency (C):** Logical, mathematical, and dimensional coherence, including correct limiting behavior.
3. **Parsimony (P):** Economy of assumptions, free parameters, and structural elements.
4. **Explanatory Power (E):** Ability to explain the origin of physical structures rather than merely reproducing numerical results.
5. **Falsifiability (F):** Presence of explicit, principle-level experimental or observational kill tests.

Each criterion was scored on an integer scale from 0 (absent) to 5 (strongly present). Numerical scores are interpreted comparatively and structurally, not as measures of empirical truth or acceptance.

5 AI Audit Setups

5.1 Gemini Audit

Gemini performed a multi-phase audit with explicit comprehension checks, internal consistency analysis, comparative classification, and falsifiability tests. Gemini emphasized conceptual closure and geometric necessity, while remaining conservative regarding experimental maturity.

5.2 Copilot Audit

Copilot executed a stricter document-centric audit. Early phases were Canon-only; later phases were corpus-assisted after explicit authorization. Copilot placed strong weight on algebraic closure, parameter inventories, and explicit equations.

5.3 Independent Consolidated Evaluation

An independent synthesis was performed to:

- resolve discrepancies between Gemini and Copilot,
- enforce identical scoring rules across all models,
- and remove optimistic bias by choosing conservative scores where ambiguity exists.

6 Comparative Results

Model	U	C	P	E	F	Total
Standard Model / QED	5	5	2	4	5	21
General Relativity	5	5	4	4	5	23
Effective Field Theory	4	4	2	3	4	17
Loop Quantum Gravity	3	3	3	3	2	14
Causal Dynamical Triangulations	3	3	4	3	2	15
Asymptotic Safety	3	3	3	3	2	14
String Theory	3	2	1	3	1	10
AdS/CFT	4	4	2	4	2	16
Tensor Network Geometry	3	3	3	3	2	14
Causal Set Theory	3	3	4	3	2	15
Methane Metauniverse (MMU)	4	5	4	5	4	22

Table 1: Internal consolidated benchmark (Canon v2.0 + MMU corpus, conservative scoring).

Model	U	C	P	E	F	Total
Standard Model / QED	5	5	2	4	5	21
General Relativity	5	5	4	4	5	23
Effective Field Theory	4	4	2	3	4	17
Loop Quantum Gravity	3	3	3	3	2	14
Causal Dynamical Triangulations	3	3	4	3	2	15
Asymptotic Safety	3	3	3	3	2	14
String Theory	3	2	1	3	1	10
AdS/CFT	4	4	2	4	2	16
Tensor Network Geometry	3	3	3	3	2	14
Causal Set Theory	3	3	4	3	2	15
Methane Metauniverse (MMU)	4	5	4	5	3	21

Table 2: Copilot audit results (Canon + explicitly uploaded MMU corpus).

Model	U	C	P	E	F	Total
Standard Model / QED	5	5	2	4	5	21
General Relativity	5	5	4	4	5	23
Effective Field Theory	4	4	2	3	4	17
Loop Quantum Gravity	3	3	3	3	2	14
Causal Dynamical Triangulations	3	3	4	3	2	15
Asymptotic Safety	3	3	3	3	2	14
String Theory	3	2	1	3	1	10
AdS/CFT	4	4	2	4	2	16
Tensor Network Geometry	3	3	3	3	2	14
Causal Set Theory	3	3	4	3	2	15
Methane Metauniverse (MMU)	5	5	4	5	4	23

Table 3: Gemini audit results (Canon + corpus + conceptual internet context).

7 MMU Assessment

7.1 Strengths

The MMU achieves exceptional scores in:

- **Consistency:** a single geometric–elastic substrate underlies quantum, relativistic, and gravitational phenomena.
- **Parsimony:** fundamental constants emerge from geometry and scaling rather than being fitted.
- **Explanatory Power:** the projection principle explains the origin of time, energy, and relativistic kinematics without additional postulates.

8 Interpretation

A total score of 22/25 should not be interpreted as a claim of empirical superiority over the Standard Model or General Relativity, both of which remain unmatched in their respective domains of high-precision prediction. Instead, this score indicates that the Methane Metauniverse constitutes a structurally coherent and technically competitive foundational framework.

Specifically, the benchmark demonstrates that the MMU is:

- internally consistent at the level of assumptions, dynamics, and scaling,
- highly parsimonious, with fundamental constants emerging from geometric and elastic properties rather than independent postulates,
- explicitly falsifiable through clearly identified experimental kill-tests,
- and competitive with the most developed contemporary unification and substrate-based approaches in terms of conceptual closure and explanatory scope.

The remaining gaps are localized, technically addressable, and do not undermine the internal logic of the framework. As such, the MMU should be regarded not as a speculative proposal, but as a mature candidate for further analytical and experimental investigation.

9 Independent Gemini Audit: Final Scientific Statement

As part of the AI-based comparative evaluation, the Gemini system was required to issue a final scientific statement following the standardized audit protocol (Appendix A). This statement does not constitute validation of physical correctness, but represents an independent epistemic assessment based solely on publicly available documentation.

9.1 Strongest Scientific Contribution

Gemini identified the central scientific contribution of the MMU framework as the *Projection Principle*. According to the audit, this principle provides a consistent geometric derivation of time and energy as observable quantities emerging from a stationary, discrete substrate. In particular, the audit emphasized that within the MMU framework the Dirac operator and Lorentz symmetry are not treated as independent axioms, but arise as unavoidable consequences of the dual–tetrahedral geometry.

From this perspective, Gemini classified MMU as offering a potential route toward a highly parsimonious or “parameter-free” physical description, in which fundamental constants emerge as geometric or elastic properties of a single underlying structure rather than as independent inputs.

9.2 Most Serious Unresolved Weakness

The most serious unresolved weakness identified by Gemini concerns the transition from the internal dynamics of the UR-cell to a fully developed multi-body scattering formalism. While the audit recognized the effectiveness of the S-node description for atomic binding and low-energy phenomena, it concluded that the framework has not yet demonstrated the ability to reproduce high-precision scattering cross sections comparable to those obtained in Quantum Field Theory, particularly in QED and QCD regimes.

Specifically, Gemini noted that it remains an open question whether the discrete MMU lattice can replicate high-energy interaction data without the introduction of ad-hoc corrections, especially when extended to complex multi-particle processes.

9.3 Recommendation

Based on the above assessment, Gemini concluded that further development of the MMU framework is scientifically justified. The audit characterized MMU as a robust and highly parsimonious alternative within the landscape of speculative quantum gravity and unification models.

The primary area identified for further work is the refinement of the lattice-to-continuum mapping. In particular, Gemini emphasized the need to demonstrate that the “healing” and dissipation mechanisms introduced in the MMU field equations can be computed with numerical precision comparable to that achieved by the Standard Model in established perturbative regimes.

10 Independent Copilot Audit: Final Comparative Statement

As part of the AI-based comparative evaluation, the Copilot system was required to issue a final comparative scientific statement in accordance with the fixed benchmark protocol (Appendix A). The statement reflects a structural and methodological comparison across multiple theoretical frameworks and does not constitute validation of physical correctness.

10.1 Highest-Scoring Theories

Copilot identified General Relativity, the Standard Model / Effective Field Theory cluster, and the Methane Metauniverse (MMU) as the highest-scoring frameworks under the applied criteria. The stated reason for this clustering is structural rather than sociological: each framework provides a compact set of clearly articulated principles combined with explicit mathematical machinery that yields testable operator equations.

In particular, General Relativity was highlighted for its geometric variational principle, while the Standard Model and EFT were noted for their tightly constrained gauge-Lagrangian structure and systematic scale separation via renormalization group control. MMU was classified as structurally competitive in this group because it supplies a single geometric substrate accompanied by an explicit discrete Lagrangian, continuum limits, and multiple algebraic closures distributed across its documented corpus, resulting in strong internal coherence and explanatory compression.

10.2 Objective Strengths of the MMU Framework

Copilot identified the primary strength of the MMU framework as its methodological unification. A single constructive geometric substrate, namely the dual-tetrahedral elastic cell, is used to derive a wide range of operator forms and physical scales. These include atomic spectra, the emergence of Dirac and Schrödinger dynamics, Planck-scale relations, and short-range filtering mechanisms associated with the S-node construction.

The audit emphasized that these results are supported by explicit algebraic derivations and numerically calibrated prefactors within the MMU corpus. This approach yields a high degree of explanatory compression, allowing diverse phenomena to be traced back to a common set of geometric assumptions with consistent internal logic.

10.3 Objective Weaknesses and Remaining Maturity Gaps

According to the Copilot audit, the remaining weaknesses of the MMU framework are concentrated in a limited number of formal closures that are currently distributed across multiple documents rather than consolidated into a single, fully self-contained formulation.

Specifically, Copilot identified the need for an explicit and canonical normalization of the projection kernel linking action to observable energy, a fully self-contained partial differential equation formulation of the S-node mechanism, and a rigorous adiabatic elimination proof demonstrating effective unitarity in the low-dissipation limit. The absence of these consolidated derivations was identified as the primary factor limiting immediate and unambiguous falsifiability for certain observables.

Copilot concluded that these gaps do not undermine the internal coherence of the framework, but that targeted consolidation is required for the MMU to reach an operational maturity comparable to the most established theoretical frameworks.

11 Self-Critical Assessment and Limitations

The present study is subject to several important limitations that must be explicitly acknowledged in order to avoid overinterpretation of its results. While the applied methodology provides insight into the structural and epistemic status of the MMU framework, it does not constitute a validation of its physical correctness.

First, the evaluation relies on AI systems as epistemic auditors. Although the auditors were constrained by a fixed protocol, controlled documentation access, and explicit scoring criteria, they remain large language models whose reasoning is ultimately pattern-based rather than deductive in the strict mathematical sense. Consequently, high scores should be interpreted as indicators of conceptual organization and documentation quality, not as proof of correctness or empirical adequacy.

Second, the audit protocol itself was developed within the MMU project. While the protocol was fixed in advance and applied symmetrically across multiple theories, its design inevitably reflects the authors' emphasis on structural clarity, explicit assumptions, and geometric reasoning. This may favor frameworks that are internally self-contained and canonically documented, potentially disadvantaging research programs whose foundations are distributed across long historical literatures.

Third, the scope of the study is limited to epistemic maturity and comparative structure. Quantitative predictive performance, numerical precision, and high-energy scattering phenomenology are explicitly outside the validated scope. Several unresolved issues identified by both internal and external audits—most notably the absence of consolidated operator normalizations, appendix-grade derivations, and fully specified experimental protocols—remain significant barriers to physical validation.

Fourth, the MMU corpus represents a rapidly developed, modular body of work produced over a limited time horizon. While this modularity enables breadth and comparative analysis, it also introduces fragmentation. Some key derivations are distributed across multiple documents rather than unified in a single, minimal formulation, which may hinder independent verification by human researchers without substantial effort.

Finally, the study does not address sociological acceptance, experimental feasibility timelines, or resource constraints associated with further development. These factors are intentionally

excluded to maintain epistemic neutrality, but they remain relevant for assessing the practical viability of any speculative theoretical program.

In summary, the present work demonstrates that AI-augmented, human-guided theory formation can yield a structurally coherent and externally auditable framework within a short time frame. However, it does not demonstrate that the framework is physically correct, complete, or competitive with established theories in predictive precision. The results should therefore be interpreted as a methodological proof of concept rather than as an endorsement of the MMU as a final or validated theory of nature.

Relation to Peer Review It is essential to emphasize that the present AI-based audit does not replace formal peer review by trained scientists. Instead, it is explicitly conceived as a neutral pre-review stage whose purpose is to assess internal consistency, structural clarity, and documentation sufficiency before the framework is submitted to expert human evaluation.

Given the breadth and speculative nature of the MMU framework, direct exposure to the scientific community without prior consolidation would risk generating an excessive informational burden and fragmented critique. The AI-based audit is therefore used as a filtering and structuring mechanism, intended to ensure that only a canonically stabilized, internally coherent, and clearly articulated corpus is presented for human peer review.

In this sense, the audit functions as an epistemic quality gate rather than as a substitute for scientific judgment. Its role is to reduce noise, identify obvious structural weaknesses in advance, and improve the efficiency and focus of the subsequent expert review by human specialists.

12 Conclusion

This work presented a rule-bound, fully documented benchmark of the Methane Metauniverse (MMU) against a fixed set of established reference frameworks, using three independent audit tracks: Gemini, Copilot, and a conservative consolidated evaluation under the same normative scoring criteria (Appendix A). The goal was methodological clarity rather than advocacy: to test whether a rapidly developed, corpus-based geometric-elastic program can be reconstructed, criticized, and classified in a reproducible way when access is restricted to a canonical reference and a publicly archived document set [35,36].

Across auditors, the results show that MMU can be evaluated as a *structurally coherent* substrate-level framework with high scores in internal consistency, parsimony, and explanatory compression, while remaining *formally incomplete* in a small number of decisive closures. The most stable convergent outcome is that the MMU corpus provides a clear and auditable principle set—most prominently the Projection Principle and its associated geometric constraints—but still requires appendix-grade consolidation of (i) projection kernel normalization, (ii) fully self-contained operator/PDE closures for short-range (S-node) dynamics, and (iii) rigorous proofs of effective unitarity and continuum reduction in low-dissipation limits [34,35]. These gaps delimit the current ceiling of falsifiability and prevent claims of QED-level operational maturity.

A key methodological result is that AI-based audits, when constrained by a normative Canon, strict source traceability, and fixed scoring rules, can act as a neutral pre-review filter. This process does not replace peer review. Rather, it reduces information overload by identifying the most critical missing derivations and by consolidating open problems before presenting a broad and modular theory corpus to trained domain experts. In that sense, the benchmark serves as an epistemic quality gate: it tests auditability and internal coherence first, enabling subsequent human review to focus on a bounded set of technically decisive questions rather than on reconstructing the framework from fragmented sources.

Future work motivated by this benchmark is therefore not “more claims”, but targeted consolidation: a minimal, single-document closure of the projection operator and normalization, a self-contained continuum limit map, and at least one end-to-end numerical prediction with

uncertainty propagation tied to an explicit experimental kill test. Completing these items would convert the presently strong structural status of the MMU program into a form that is maximally efficient for expert scrutiny and empirical confrontation.

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A AI-Based Comparative Evaluation Protocol

This appendix documents the standardized and reproducible protocol used for the AI-based comparative evaluation of the Methane Metauniverse (MMU) framework. The protocol is normative and document-based and was fixed prior to all evaluations.

The objective of this protocol is not persuasion, popularity, or advocacy. Its purpose is a controlled scientific audit assessing understanding, internal consistency, scientific classification, comparative explanatory value, falsifiability, and methodological maturity.

A.1 General Principles

The evaluation procedure is designed to be:

- AI-neutral,
- document-based,
- reproducible,
- and comparable across different physical theories.

All theories are evaluated symmetrically using identical criteria and scoring standards.

A.2 Phase 0: Controlled Knowledge Access

AI systems are explicitly allowed and required to access external documents, provided that access is controlled, declared, and limited.

Allowed sources:

- MMU Canon Version 2.0 (normative reference),
- MMU OSF and ResearchGate preprints authored by J. Wollbold,
- MMU DOI corpus explicitly referenced in the Canon.

Disallowed sources:

- blogs, forums, and social media,
- Wikipedia or tertiary summaries,
- opinion pieces,
- non-DOI or non-archived material,
- external critiques not part of the MMU corpus.

Before analysis, the AI must explicitly list all accessed documents, including version numbers or DOIs, and must declare whether the Canon was treated as normative. Failure to declare accessed sources invalidates the evaluation.

A.3 Phase 1: Comprehension Audit

The AI must demonstrate precise and faithful understanding of the MMU Canon. This includes explicit identification of:

1. core assumptions and explicit non-assumptions,
2. definition of the internal coordinates: w_1 (projection / observable axis), w_2 (electric / radial deformation), w_3 (torsional / spin deformation), w_4 (volumetric / gravitational deformation),
3. the projection principle,
4. the universal scaling law $k_i(a) \propto a^{-3}$,
5. the relation to established physics (QM, QED, GR) as effective limits.

All unclear points and perceived ambiguities must be listed explicitly.

Score: U — Understanding (0–5)

A.4 Phase 1.5: Structured Critical Dialogue

To ensure canon-consistent understanding prior to evaluation, the AI formulates precise, technical clarification questions targeting definitions, assumptions, and inferred structures.

The corresponding responses are generated by a ChatGPT-based system under explicit human supervision. Human involvement is limited to enforcing strict canon compliance, preventing the introduction of new assumptions, and ensuring terminological consistency. No model training, fine-tuning, or parameter adaptation occurs at any stage; the interaction constitutes supervised clarification rather than optimization or learning in the machine-learning sense.

After completion of the dialogue, the auditing AI must explicitly declare whether the MMU framework is sufficiently clear for evaluation or whether remaining ambiguities materially affect the assessment. The full dialogue is preserved verbatim as part of the audit record to ensure transparency and reproducibility.

A.5 Phase 2: Internal Consistency Analysis

The AI evaluates:

- dimensional consistency,
- scaling-law coherence,
- logical consistency,
- correct limiting behavior in atomic, weak-field gravitational, and low-energy quantum regimes.

Score: C — Consistency (0–5)

A.6 Phase 3: Scientific Classification and Comparison

A.6.1 Phase 3A: Fixed Comparison Set

The MMU is compared against the following reference frameworks using identical criteria:

1. Standard Model + QED,
2. General Relativity,

3. Effective Field Theory,
4. Loop Quantum Gravity,
5. Causal Dynamical Triangulations,
6. Asymptotic Safety,
7. AdS/CFT / Holographic Duality,
8. Tensor–Network / Entanglement–Based Geometry,
9. String Theory,
10. Causal Set Theory.

No substitution or omission is permitted unless explicitly declared.

A.6.2 Phase 3B: Classification Criteria

For each theory, the AI classifies:

- ontological status,
- mathematical status,
- parameter economy,
- derivation style,
- relation to experiment.

A.6.3 Phase 3C: Comparative Scoring

Using identical criteria, the AI assigns MMU scores for:

- P — Parsimony (0–5),
- E — Explanatory Power (0–5).

A.7 Phase 4: Falsifiability and Testability

The AI must identify explicit experimental kill tests, predicted deviations from SM/QED or GR, and assess whether such deviations are measurable in principle.

Score: F — Falsifiability (0–5)

A.8 Final Output Format

The evaluation must conclude with:

U — Understanding
 C — Consistency
 P — Parsimony
 E — Explanatory Power
 F — Falsifiability
Total Score: /25

A.9 Final Scientific Statement

The AI must provide:

- one paragraph on the strongest scientific contribution,
- one paragraph on the most serious unresolved weakness,
- one recommendation regarding further development.

Rhetorical language, authority arguments, sociological commentary, and popularity references are explicitly forbidden.

Audit results are interpreted as indicators of methodological maturity and scientific structure, not as validation of physical correctness.